

# SONG FEI

PhD Student @ HKUST(GZ) · AIGC & Low Level Vision

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## RESEARCH SUMMARY

PhD student specializing in **Generative AI** and **Low Level Vision**. I build systems that define the **State-of-the-Art**:

- **SOTA Visual Frontiers**: Pioneered **LucidFlux** and **UltraFlux**, pushing visual generation quality beyond strong commercial baselines.

## EDUCATION

**The Hong Kong University of Science and Technology, Guangzhou**

PhD Student

Guangzhou, China

Aug. 2026 – Present

- Advisor: Prof. Lei Zhu
- Research Focus: Text-to-Image Generation, Visual Restoration.

**The Hong Kong University of Science and Technology, Guangzhou**

MPhil in Robotics and Autonomous Systems (ROAS)

Guangzhou, China

Aug. 2024 – Aug. 2026

- Advisor: Prof. Lei Zhu and Prof. Jun Ma
- Research Focus: Text-to-Image Generation, Visual Restoration.

**Northeastern University**

B.Eng. in Industrial Intelligence

Shenyang, China

Sep. 2020 – Jul. 2024

## RESEARCH EXPERIENCE

**JD Retail**

TGT (Tech Genius Team) Research Intern

Beijing, China

Jun. 2026 – Present

- **E-commerce Video Generation**: Worked on post-training for multi-reference image-conditioned and audio-driven video generation, improving product fidelity, temporal consistency, and audio-visual synchronization for retail content creation.

**OPPO Research Institute**

Research Scientist Intern (Image Super-Resolution & Restoration)

Shenzhen, China

Dec. 2025 – Present

- **Image Restoration & Super-Resolution**: Contributed to the research and development of advanced generative models for image super-resolution and degradation restoration, targeting real-world mobile photography scenarios. The proposed methods improved perceptual quality and detail reconstruction under low-light and compressed conditions.

**The Hong Kong University of Science and Technology, Guangzhou**

MPhil Student

Guangzhou, China

Aug. 2024 – Aug. 2026

- **Visual Frontiers**: Served as a core contributor to **LucidFlux** (ICLR'26) and **UltraFlux** (CVPR'26). LucidFlux pushes the boundary of visual quality, surpassing strong commercial baselines (e.g., **Meitu SR**), and has been successfully launched as an **Official API on Fal.AI**.

## SELECTED PUBLICATIONS

**Summary**: Published 7+ papers in top-tier conferences/journals (CVPR, ICCV, ICLR, MICCAI), including 3 first-author/co-first-author papers.

\* *Equal Contribution*.

### First-Author & Co-First-Author Publications

**[CVPR 2026 Highlight] UltraFlux: Data-Model Co-Design for High-quality Native 4K Text-to-Image Generation across Diverse Aspect Ratios**

Tian Ye\*, **Song Fei\***, Lei Zhu.

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

**[ICLR 2026] LucidFlux: Caption-Free Photo-Realistic Image Restoration via a Large-Scale Diffusion Transformer**

**Song Fei\***, Tian Ye\*, Lujia Wang, Lei Zhu.

*International Conference on Learning Representations (ICLR)*, 2026.

**[MICCAI 2026] S2DLight: Single Step Diffusion Network via Degradation-Aware Adaptation for Surgical Endoscopic Image Low-Light Enhancement**

**Song Fei**, Guanyi Wang, Tian Ye, Sixiang Chen, Jianyu Lai, Lei Zhu.

*International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.*

**[NeurIPS 2026 Under Review] LucidNFT: LR-Anchored Multi-Reward Preference Optimization for Flow-Based Real-World Super-Resolution**

**Song Fei\***, Tian Ye\*, Sixiang Chen, Zhaohu Xing, Jianyu Lai, Lei Zhu.

*Neural Information Processing Systems (NeurIPS), 2026.*

**Co-Authored Publications**

**[MICCAI 2026] MoSE: Mixture-of-Scale-Experts for Medical Image Restoration**

Xingyu Li, Zhaohu Xing, Zekai Liu, Shuaibo Li, Hongqiu Wang, **Song Fei**, Lei Zhu.

*International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.*

**[ICLR 2026] PosterCraft: Rethinking High-Quality Aesthetic Poster Generation in a Unified Framework**

SiXiang Chen\*, Jianyu Lai\*, Jialin Gao\*, Tian Ye, Haoyu Chen, Hengyu Shi, Shitong Shao, Yunlong Lin, **Song Fei**, Zhaohu Xing, Yeying Jin, Junfeng Luo, Xiaoming Wei, Lei Zhu.

*International Conference on Learning Representations (ICLR), 2026.*

**[CVPR 2025] SnowMaster: Comprehensive Real-world Image Desnowing via MLLM**

Jianyu Lai, Sixiang Chen, Yunlong Lin, Tian Ye, Yun Liu, **Song Fei**, Zhaohu Xing, Hongtao Wu, Weiming Wang, Lei Zhu.

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.*

**[ICCV 2025] GenHaze: Pioneering Controllable One-Step Realistic Haze Generation for Real-World Dehazing**

Sixiang Chen, Tian Ye, Yunlong Lin, Yeying Jin, Yijun Yang, Haoyu Chen, Jianyu Lai, **Song Fei**, Zhaohu Xing, Fugee Tsung, Lei Zhu.

*International Conference on Computer Vision (ICCV), 2025.*

**HONORS & ACADEMIC SERVICE**

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**Honors & Awards**

- Postgraduate Scholarship of HKUST(GZ), 2024

**Academic Service**

- **Conference Reviewer:** AAAI.